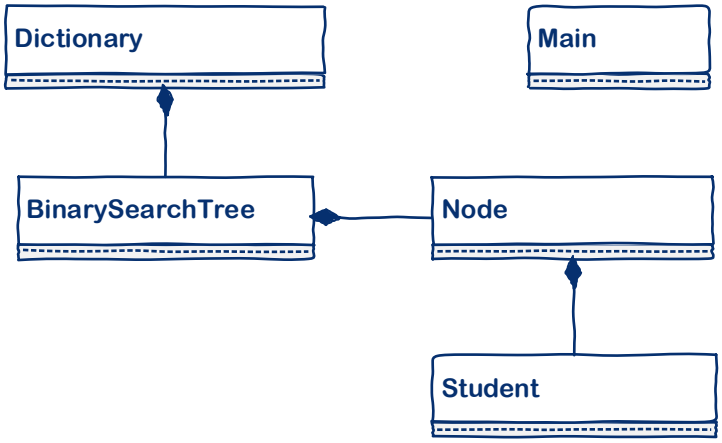
COP3530 – Assignment 3

Objective

Students will practice the use of binary search trees; students will be able to develop a solution to a problem involving the implementation of a dictionary abstract data type using a binary search tree data structure.

Assignment Problem

In a Dictionary ADT each record of information contains two parts, a key and the value associated with that key. In this assignment you are asked to implement a dictionary with the operations described below and the data structure of choice is a binary search tree. UML class diagram:



Guidelines

- The assignment is to be completed individually or in teams of two students. Questions are based on the content studied in class on the Dictionary ADT and the binary search tree.

- Solution to the assignment is expected to be based on code given in programming Exercise 22_01 (found in Canvas) and solution to related exercises developed in class. Please make sure you properly credit its source.

- Classes from the Java Collection Framework, are not allowed for use in your implementation. This is a strong requirement, if you have questions, don't hesitate to ask.

- Students are required to structure the code as indicated in the UML class diagram above.

- Students are required to develop implementations according to the instructions provided in the code accompanying this assignment.

- Several files accompany this project description:
    - *assignment 3 test set.txt*: a test file,
    - *Main.java*: a complete implementation of the Main class (tests your code using file *assignment 3 test set.txt*),
    - *Dictionary.java*: the framework of the Dictionary class, with the constructor and toString methods implemented,
    - *BinarySearchTree.java*: the framework of the BinarySearchTree class, with the constructor and the toString methods implemented,
    - *Node.java, Student.java*: implementations of the Node and Student classes, respectively,
    - *student database.csv*: a sample dictionary
    - *output.pdf*: a file with the output generated by the Main class provided.

- Students are required to use the given class frameworks included, completing all the methods as indicated in the Javadoc comment section of each method.

- The toString methods provided must be used in all output. Students are not allowed to modify it or use other form of output.

- The output of your program is required to be the same as the output example provided.

Deliverables

• A compressed folder, *PID Assignment 3 (for example, 1234567 Assignment 2)*, containing the files:

- *Dictionary.java*,
- *BinarySearchTree.java*,
- A screenshot of the running program. Note: a screenshot is an image that shows 1) the IDE environment with code and 2) the output window. More than one image file might be submitted.

• Include **only** the .java files mentioned above; do not include other files or folders generated by the IDE.

• Project submission containing .class files instead of .java files will be considered unsubmitted and will be given a grade of 0.

• Make sure you write the full name, class section, and Panther ID of each author in the @author field of the Java classes.

• Team submission: each team member must enter all names, class sections, and PIDs of the team in the comment window of the project dropbox. Select a team leader to submit the project (only team leaders submit the project, but the info of the team members must be entered by each student in the team).

Grading Rubric

The assignment is worth 145 points (out of 1000 total course points). Grade components:

<table>
<tr><th>Component</th><th>Points</th><th>Description</th></tr>
<tr><td>Submission</td><td>5</td><td>The student has submitted the project solution using the requirements for deliverables specified in the Deliverables section.</td></tr>
<tr><td>Organization</td><td>5</td><td>Code is expected to be neat, organized, and readable.</td></tr>
<tr><td>Content</td><td>135</td><td>
<table>
<tr><th>Element</th><th>Points</th></tr>
<tr><td>Dictionary.java</td><td>40 pts</td></tr>
<tr><td>BinarySearchTree.java</td><td>90 pts</td></tr>
<tr><td>Screenshot(s)</td><td>5</td></tr>
</table>
</td></tr>
</table>